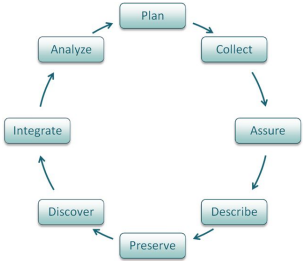
Big Data to Knowledge(BD2K)

Best Practices for Biomedical Research Data Management

Module 1

Introduction and Overview



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Learning Objectives

1. Describe the wide variety of types of research data
2. Describe the research data life cycle
3. Explain why managing and sharing research data is important
4. Identify intent of data policies that have been emerging
5. Identify major government funding agencies currently requiring a data management plan
6. Describe the typical categories of information needed to complete a data management plan
7. Understand the support role an information professional (librarian) can play in research data management planning

Research Data Can Take Many Forms

Social Science Data:

- Survey responses
- Focus groups
- Individual interviews
- Economic indicators
- Demographics
- Opinion polling

Hard Science Data:

- Measurements generated by sensors/laboratory instruments
- Computer modeling
- Simulations
- Observations and/or field studies
- Specimen

Data Forms Used By Both:

- Images
- Video
- Mapping/GIS data
- Numerical measurements

Research Data Types

Qualitative: non-numerical, unstructured data

Quantitative: numerical data, structured data

**More and more studies are using mixed methods with both quantitative and qualitative data being collected

Research Data Categories

- Collecting or Creating Raw Data
- Processing Data
- Analyzing Data
- Finalizing/Publishing Data
- Preserving or Archiving Data
- Sharing Data
- Re-Using Data

Research Data Life Cycle

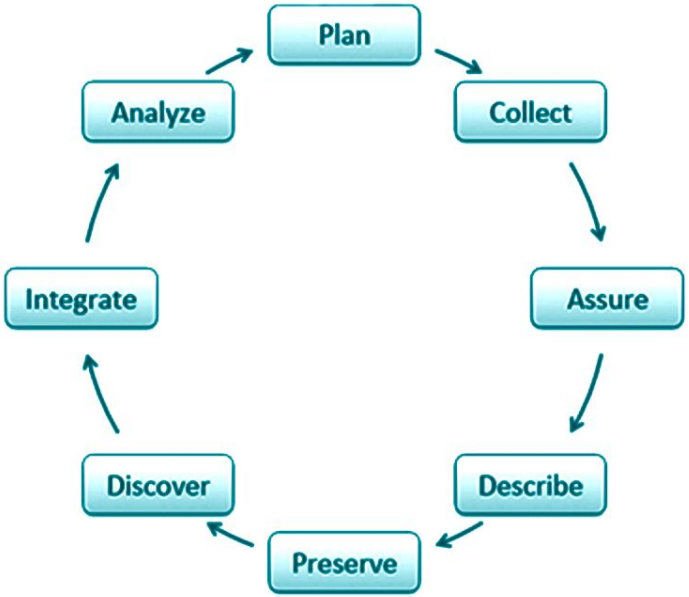

Image credit: Dataone Leadership Team
https://www.dataone.org/data-life-cycle

Data Policies

- Data ownership and intellectual property
- Data management and stewardship responsibilities
- Public access, data sharing & dissemination
- Retention

U.S.Department of Health & Human Services

NIH Genomic Data Sharing (GDS)

Federal Policies

February 2013: Office of Science and Technology Policy at the White House issued a directive that Federal Agencies with more than $100 million in Research & Development design plans to make the research of federally funded research freely available to the public

May 2013: President Obama issued an Executive Order "Making Open and Machine Readable the New Default for Government Information"

**For the complete history and current direction of Federal Open Access Policies see SPARC's Data Sharing Requirements by Federal Agency http://datasharing.sparcopen.org

NSF Data Management and Sharing Plans

- The types of data, samples, physical collections, software, curriculum materials, and other materials to be produced in the course of the project
- The standards to be used for data and metadata format and content (where existing standards are absent or deemed inadequate, this should be documented along with any proposed solutions or remedies)
- Policies for access and sharing including provisions for appropriate protection of privacy, confidentiality, security, intellectual property, or other rights or requirements
- Policies and provisions for re-use, re-distribution, and the production of derivatives
- Plans for archiving data, samples, and other research products, and for preservation of access to them